MAT244 (ODEs) Notes

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1 Introduction to Ordinary Differential Equations

This lecture corresponds to §1.1-1.3 in the textbook.

1.1 What Are Differential Equations?

An **differential equation**, which I will often abreviate as a **DE**, is an equation containing derivatives of an unknown function. The two main types of DEs are **ordinary differential equations**, or **ODEs**, which are equations involving functions depending on a single independent variable, and **partial differential equations**, or **PDEs**, which are equations involving functions which depend on more than one independent variable. For example, the equation:

$$\frac{d^2u}{dt^2} + \frac{du}{dt} + u = f(t)$$

is an ODE that depends on the independent variable t. The equation:

$$-\partial_t^2 u + \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$$

is an example of a PDE that depends on multiple independent variables x, y, z, and t. The focus of this course will be on ODEs.

Example 1.1.1. Consider the following ODE:

$$\frac{dy}{dt} = -y(t) + t$$

How can we determine what this equation represents? From calculus, we know that dy/dt is the derivative of $y(t)$, and that this derivative represents how fast the function y is changing with respect to t. Since we have that $dy/dt = -y(t) + t$, the right hand side shows that the slope will decrease as y increases and will increase as t increases.

Recall that $\left.\frac{dy}{dt}\right|_p$ represents the slope of $y(t)$ passing through the point p. We can thus say that the **direction** of $y(t)$ at point $t = t_0$ is equal to $-y(t_0) + t_0$. With this information, we can plot a **direction field** (or **slope field**) for this differential equation:

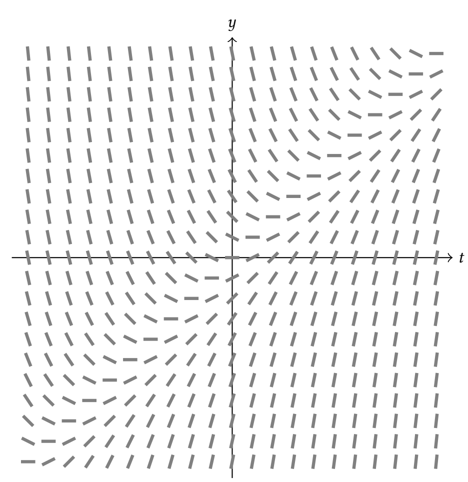

Figure 1: The slope field for the equation $\frac{dy}{dt} = -y(t) + t$

Remark that, in accordance with our intuition, the slope decreases as we move up the y axis and it increases as we move up the t axis. Each line segment in Figure 1 represents the slope of the equation at one of its solutions. Note that if we travel along the line $y(t) = t$, we see that the slope of the solution is always zero. We say that the line $y(t) = t$ is the **equilibrium solution** of the DE and that it does not change with respect to time (i.e. $\frac{dy}{dt} = 0$):

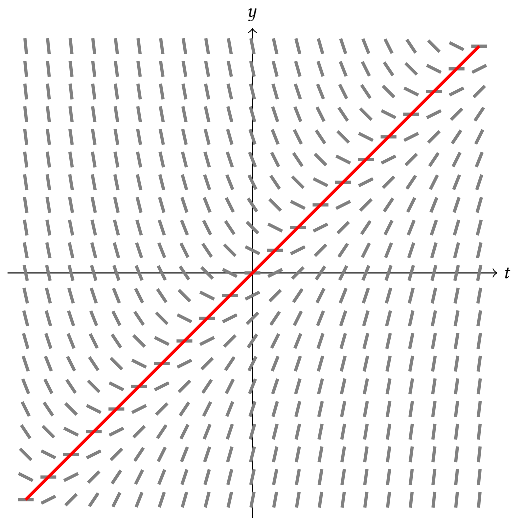

Figure 2: The equilibrium solution (in red) to the equation $\frac{dy}{dt} = -y(t) + t$ is $y(t) = t$.

Example 1.1.2. Consider the differential equation $\frac{dy}{dt} = ay$, where $a \in \mathbb{R}$. Plot direction fields for $a = -1,0$, and 1. What is/are the equilibrium solution(s)?

Solution. When $a = -1$, we have that $\frac{dy}{dt} = -y$, i.e. the slope of the equation is equal to $-y$. Observe that $\frac{dy}{dt}$ does not change when t changes. The direction field is as follows:

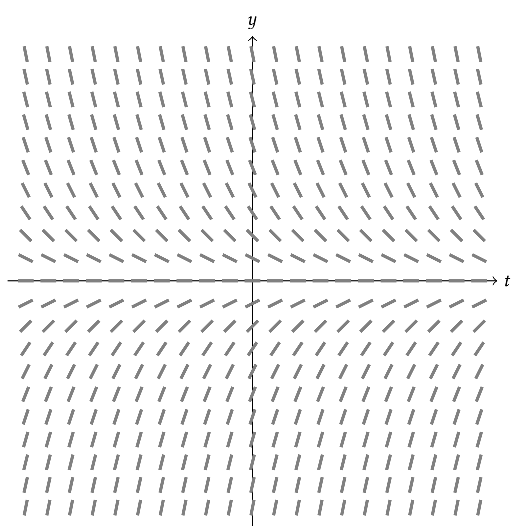

Figure 3: The direction field for $\frac{dy}{dt} = -y$.

Observe that the slope of the equation is equal to the value of $-y$. For example, when $y = 1$, we have that $\frac{dy}{dt} = -1$. Furthermore note that $\frac{dy}{dt}$ is independent of t and hence does not change as t changes. The equilibrium solution for this system is the x-axis $y = 0$.

The direction field for $a = 1$, i.e. $\frac{dy}{dt} = y$, can be plotted in a very similar way.

When $a = 0$, we have that $\frac{dy}{dt} = 0$. Hence the slopes in the direction field will equal zero for all t and y:

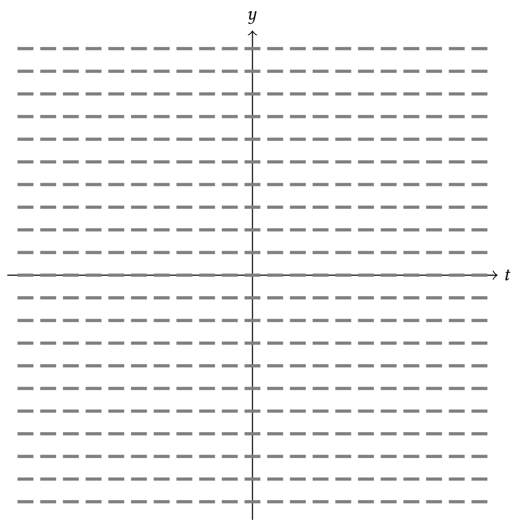

Figure 4: The direction field for $\frac{dy}{dt} = 0$.

Since the slope is always zero, any line passing through the direction field will have slope zero. There are hence infinitely many equilibrium solutions. ✓

1.2 Solutions of Differential Equations

We "solve" differential equations of the form $dy/dt = f(t, y)$ by finding a function $y(t)$ that satisfies it.

Example 1.2.3. Solve the differential equation $\frac{dy}{dx} = 2y(t)$.

Solution. First, rewrite the equation so that all the terms with y appear on one side of the equation and all the terms with t appear on the other:

$$\frac{dy}{dt} = 2y \implies \frac{dy}{y} = 2dt$$

Though it may seem bizarre that we separated dy/dt into two components, we are allowed to do so when solving DEs. We will provide more justified reasoning for doing so in the next lecture.

Now we integrate both sides:

$$\int \frac{dy}{y} = \int 2dt$$
$$\ln|y| = 2t + C$$

Now solve for y:

$$\begin{aligned} \ln|y| &= 2t + C \\ |y| &= e^{2t+C} \\ y(t) &= \pm e^{2t} e^{C} \end{aligned}$$

Note that e^C is a positive constant, so we can replace it with a new constant $A = e^C$ such that:

$$y(t) = Ae^{2t}$$

Lastly let us check our work by integrating $y(t)$ and ensuring it matches the given equation:

$$\begin{aligned} y' &= 2Ae^{2t} \\ y' &= 2y(t) \end{aligned} \qquad \checkmark$$

Suppose that we are also given an initial condition, for instance suppose for the previous example that $y(0) = 1$. This initial condition uniquely determines the value of the constant and hence the function y. With the above example, if we assume $y(0) = 1$, then we find that:

$$\begin{aligned} y(0) &= 1 = Ae^{2(0)} \\ 1 &= A \end{aligned}$$

Example 1.2.4. Solve the ODE $\frac{dy}{dx} = 3y - 2$, given the initial condition $y(0) = 2$.

Solution. First, rearrange the equation so that all the terms with y are on one side and all the terms with x are on the other (remember we are allowed to split $\frac{dy}{dx}$ into components):

$$\frac{1}{3y-2} dy = dx$$

Now integrate both sides:

$$\begin{aligned} \int \frac{1}{3y-2} dy &= \int dx \\ \frac{1}{3}\ln|3y-2| + C_1 &= x + C_2 \end{aligned}$$

If we put the constants together we obtain:

$$\begin{aligned} \ln|3y-2| &= 3x + C \\ |3y-2| &= Ae^{3x} \\ 3y - 2 &= \pm Ae^{3x} \\ 3y &= \pm Ae^{3x} + 2 \\ y &= \frac{\pm Ae^{3x} + 2}{3} \end{aligned}$$

Now solving for the initial value:

$$
\begin{aligned}
y(0) = 2 &= \frac{\pm A e^{3(0)} + 2}{3} \\
2 &= \frac{A + 2}{3} \\
6 &= A + 2 \implies A = 4
\end{aligned}
$$

So a solution to the initial value problem is $y(t) = \frac{4e^{3t}+2}{3}$. ✓

1.3 Classification of Differential Equations

The **order** of a DE is the highest derivative in the equation. For example, the order of $y' = y(t)$ is 1, and the order of $y'' + y + t = 0$ is 2. The general form of an order n DE is:

$$F(x, u(x), u'(x), u''(x), \ldots, u^{(n)}(x)) = 0$$

A solution of the ODE $y^{(n)} = f(x, y, y', \ldots, y^{(n-1)})$ on the interval $a < x < b$ is a function ϕ such that $\phi', \phi'', \ldots, \phi^{(n)}$ exists and satisfies $\phi^{(n)} = f(x, \phi, \phi', \ldots, \phi^{(n-1)})$ for all $x \in (a, b)$. Note that ϕ and all necessary derivatives must exist on $a < x < b$ and ϕ must satisfy the differential equation.

For example, consider $y'' + y = 0$, or $y'' = -y$. Solutions to this DE will be of the form e^{ix} or make use of sines and cosines (for instance, it is easy to check that $y = \sin(x) + \cos(x)$ is a solution). However, naturally the question arises: does a solution always exist, if so it is unique? And how do we find it? We will learn how to answer those questions in the next lecture.

2 Linear & Nonlinear ODEs, Integrating Factors

This lecture corresponds to §1.3 and §2.1 in the textbook.

2.1 Linear & Nonlinear ODEs

Definition 2.1.1

We say a differential equation $F(x, y, y', y'', \ldots, y^{(n)}) = 0$ is **linear** if F is a linear function of $y, y', y'', \ldots, y^{(n)}$. Thus the general form of a linear differential equation is:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

We say an equation is **nonlinear** if it is not linear.

The equation $y + y'' = 0$ is an example of a linear DE, as it can be written in the form $a_0(x)y + a_2(x)y'' = 0$ where $a_0(x) = a_2(x) = 1$. Another example of a linear DE is $y' \sin(x) = x$, where $a_1(x) = \sin(x)$ and $g(x) = x$. Conversely, the equation $y' + y^2 = 0$ is a nonlinear DE as it cannot be written in a linear form.

In mathematics, linear theory has been very well developed, but nonlinear theory has not been developed to the same degree. However, we have been able to approximate nonlinear equations by linear ones using **linear approximations**.

For example, consider the following differential equation representing an oscillating pendulum:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$$

This equation is obviously nonlinear due to the $\sin\theta$. However, when the angle θ is small, we can approximate $\sin\theta \approx \theta$, hence we can linearize the equation as:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$$

which is an equation we can solve (see exercises).

2.2 Integrating Factors with Linear First Order ODEs

In this section, we will be working with equations of the form $\frac{dy}{dx} + p(x)y = g(x)$, where p and g are functions of the independent variable t. We ,introduce a new method of solutions called integrating factors, with $\mu(t)$ being the integrating factor.

Example 2.2.2. Solve the differential equation $y' + 2y = 1$ using integrating factors.

Solution. The first step in solving an ODE with integrating factors is to multiply the entire equation by a function $\mu(t)$ which we will choose so the entire function is readily integrable. We have:

$$\mu(t)\frac{dy}{dt} + \mu(t)2y = \mu(t) \tag{1}$$

The second step is to try and find the value of $\mu(t)$ such that this equation becomes readily integrable. We do this by considering the left side of Equation (1) as the derivative of some recognizable expression. Observe that the term $\mu(t)\frac{dy}{dx}$ is part of the differentiating product $\mu(t)y$. Indeed if we recall the product rule:

$$\frac{d}{dt}\left[\mu(t)y\right] = \mu(t)\frac{dy}{dt} + \frac{d\mu}{dt}y$$

we can find the integrating factor $\mu(t)$ by setting $\mu(t)2y = \frac{d\mu}{dt}y$. Isolating $\mu(t)$ we obtain:

$$\begin{aligned}
&\frac{1}{\mu(t)}d\mu = 2dt \\
\implies &\int \frac{1}{\mu(t)}d\mu = \int 2dt \\
\implies &\ln|\mu(t)| + 2t + C \\
\implies &\mu(t) = e^{2t+C} \\
\implies &\mu(t) = Ae^{2t}
\end{aligned}$$

Hence Equation (1) becomes (we can cancel out the constant A):

$$e^{2t}\frac{dy}{dt} + e^{2t}2y = e^{2t} \tag{2}$$

We remark that now the left side of Equation (2) is obviously the derivative of the product $e^{2t}y$, so we can simplify the leftside furthermore to obtain:

$$\frac{d}{dt}\left[e^{2t}y\right] = e^2t$$

The third step is to integrate both sides of the Equation (3), which we can now do with the chosen integrating factors, and isolate y. Remark that on the left side of the equation the derivative and integral cancel each other out:

$$\begin{aligned}
&\cancel{\int}\frac{\cancel{d}}{dt}\left[e^{2t}y\right]\cancel{dt} = \int e^{2t}dt \\
\implies &e^{2t}y = \frac{e^2t}{2} + C \\
\implies &y = \frac{1}{2} + Ce^{-2t} \qquad \checkmark
\end{aligned}$$

Remark. For ODEs of the form $\frac{dy}{dt} + p(t)y = g(t)$, the integrating factor is always of the form $\mu(t) = \exp\int p(t)dt$.

Example 2.2.3. Solve the differential equation $y' + 2ty = e^{-t^2}$ using integrating factors.

Solution. We begin by multiplying the equation by the integrating factor $\mu(t)$:

$$\mu(t)y' + \mu(t)2ty = \mu(t)e^{-t^2} \tag{3}$$

We first note that $p(t) = 2t$ and $g(t) = e^{-x^2}$. Using the remark above, we find that:

$$\begin{aligned} & \mu(t) = \exp \int 2t dt \\ \iff & \mu(t) = e^{t^2+C} \\ \iff & \mu(t) = Ae^{t^2} \end{aligned}$$

So Equation (3) becomes:

$$e^{t^2}\frac{dy}{dt} + 2te^{t^2}y = 1$$

We recognize the left side of the equation as the derivative of the product $\mu(t)y$. Integrating both sides of the equation yields:

$$\begin{aligned} & \cancel{\int} \frac{\cancel{d}}{dt}\left[e^{t^2}y\right]\cancel{dt} = \int dt \\ \implies & e^{t^2}y = t + C \\ \implies & y = e^{-t^2}(t + C) \end{aligned}$$

✓